① δ_a^b invariant for S. → rotations
→ lorentz group. ⇒ generic

② suppose $\exists$ an invariant made of 2 vectors.

$\vec{r}\cdot\vec{r}$, $-t^2+x^2+y^2+z^2$

$$X^\mu = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}, \quad \underbrace{\eta_{\mu\nu}}_{} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}$$

metric tensor for the Lorentz transformations.

$$X^\mu \eta_{\mu\nu} X^\nu$$

in the silly coordinates

$$\underbrace{x^2+f^2z^2}$$

$$r^i = \begin{pmatrix} x \\ z \end{pmatrix}$$

$r^1 = x$

$r^2 = z$

$$r^i g_{ij} r^j$$

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & +f^2 \end{pmatrix}$$

metric tensor.

The inverse $\eta^{\mu\nu}$ satisfies $\eta_{\mu\nu}\eta^{\mu\nu} = \delta^\mu{}_\nu$

Hence $\eta_{\mu\nu}\eta^{\mu\nu} = I$, hence

$\eta^{\mu\nu} = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$ as well.

$$g_{ij} = (R^{-T}_{silly})_i{}^m (R^{-T}_{silly})_j{}^n \, g_{mn}.$$

where: $$R_{silly} = \begin{pmatrix} \cos\theta & f\sin\theta \\ -\frac{1}{f}\sin\theta & \cos\theta \end{pmatrix}$$

$$R^{-T}_{silly} = \begin{pmatrix} \cos\theta & \frac{1}{f}\sin\theta \\ -f\sin\theta & \cos\theta \end{pmatrix}$$

$$g_{ij} = (R^{-T})_i{}^m g_{mn} (R^{-1})^n{}_j$$

$$\begin{pmatrix} c & \frac{s}{f} \\ -fs & c \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & f^2 \end{pmatrix} \begin{pmatrix} c & -fs \\ \frac{s}{f} & c \end{pmatrix} = \begin{pmatrix} c & \frac{s}{f} \\ -fs & c \end{pmatrix} \begin{pmatrix} c & -fs \\ fs & f^2 c \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & f^2 \end{pmatrix}$$

verified that g_{ij} is an invariant tensor

the metric tensor of the rotated coordinate.

$$x^\mu \eta_{\mu\nu} x^\nu = x'^\mu \eta_{\mu\nu} x'^\nu$$

matrix form.

$$x^T \eta x = (S x')^T \eta (S x')$$

$$\eta = S^T \eta' S$$

$$S^{-T} \eta S^{-1} = \eta' \rightarrow$$ this is how lower indices transform.

what are we trying to show here.

$$\eta_{\mu\nu} = (S^{-T})_\mu{}^\rho \eta_{\rho\sigma} (S^{-1})^\sigma{}_\nu$$

$\to$ row

why not $(S^{-T})^\nu{}_\sigma$? I guess you keep σ up for the convention of summation?

$$= (S^{-T})_\mu{}^\rho (S^{-T})_\nu{}^\sigma \eta_{\rho\sigma}$$

$\to$ column

$$= \eta'_{\rho\sigma} =$$

this is just how the lower indices transform.

$$(\eta_{\mu\nu} x^\nu)' = \eta'_{\mu\nu} x'^\nu = (S^{-T})_\mu{}^\rho (S^{-T})_\nu{}^\sigma \eta_{\rho\sigma} S^\nu{}_\xi x^\xi$$

$$= (S^{-T})_\mu{}^\rho \underbrace{(S^{-1})^\sigma{}_\nu S^\nu{}_\xi}_{\delta^\sigma{}_\xi} \eta_{\rho\sigma} x^\xi$$

$$= (S^{-T})_\mu{}^\rho \eta_{\rho\sigma} x^\sigma$$

$$=$$

a lower index tensor

$$x^\mu = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}, \quad x_\mu = \eta_{\mu\nu} x^\nu = \begin{pmatrix} -t \\ x \\ y \\ z \end{pmatrix}$$

this is how you lower an index.

$$(\eta^{-1})^{\rho\mu} x_\mu = (\eta^{-1})^{\rho\mu} \eta_{\mu\nu} x^\nu$$

$$= \delta^\rho{}_\nu x^\nu$$

$$= x^\rho.$$

$$(\eta^{-1})^{\rho\nu} = \eta^{\rho\nu}$$

Since we chose units:

$$\eta^{\mu\nu} = \eta_{\mu\nu}$$

but if we chose $c \neq 1$, then

$$\eta_{\mu\nu} = \begin{pmatrix} -c^2 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$$\eta^{\mu\nu} = \begin{pmatrix} -\frac{1}{c^2} & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

Tensor in Regular physics

$\sigma_{ij} C_{ijkl} \epsilon_{kl}$

stress — strain — elasticity tensor

$$t_\delta{}^{\beta\gamma} = \eta_{\delta\alpha} t^{\alpha\beta\gamma}$$

$$t_\delta{}^{\beta}{}_{\gamma} = \eta_{\delta\alpha} \eta_{\varepsilon\gamma} t^{\alpha\beta\gamma}$$

← The Levi-Civita symbol

(3) $\epsilon^{abc} = \begin{cases} \epsilon^{123} = 1 \\ \text{maximumly antisymmetric} : \text{any pair exchange changes sign} \\ \epsilon^{aba} = 0 \end{cases}$

rotations

The symbol is the antisymmetric object that encodes orientation, cross products, and volume.

ϵ^{abc} $a,b,c \in \{1,2,3\}$

1) $\epsilon^{212} = -\epsilon^{221} = 0$

2) $\epsilon^{213} = -1$

- Only 1 independent components

↳ given ϵ^{123}, then you know ϵ^{213} ϵ^{312} ϵ^{321} ....

trying to show that this is invariant under the group of rotations

the group of rotations

- Proof that $\epsilon^{123} = 1$

$$\epsilon^{123} = R^1{}_a R^2{}_b R^3{}_c \epsilon^{abc}$$

$$\det\begin{pmatrix} R^1{}_1 & R^1{}_2 & R^1{}_3 \\ R^2{}_1 & R^2{}_2 & R^2{}_3 \\ R^3{}_1 & R^3{}_2 & R^3{}_3 \end{pmatrix}$$

for any a, b, c the one 7 comb. that gives 0.

$$= R^1{}_1 R^2{}_2 R^3{}_3 - R^1{}_1 R^2{}_3 R^3{}_2 + R^1{}_2 R^2{}_3 R^3{}_1$$

$$- R^1{}_2 R^2{}_1 R^3{}_3 + R^1{}_3 R^2{}_1 R^3{}_2 - R^1{}_3 R^2{}_2 R^3{}_1$$

$$= \det(R) = 1$$

only 6 nonzero components.

$6 = 3 \times 2 \times 1$

$\epsilon^{123} = 1$

$\epsilon^{132} = -1$!

ϵ

a	b	c
1	2	3
1	3	2
2	1	3
2	3	1
3	1	2
3	2	1

$$\epsilon^{\alpha\beta\gamma\delta} = \begin{cases} \epsilon^{0123} = 1 \\ \text{maximally } \cancel{\text{antisym}} \text{ antisymmetric} \end{cases}$$

the only nonzero component must have all 1,2,3 & one three's repeated index ($\epsilon_{aab}=0$)

only $4\times3\times2\times1=24$ components that are nonzero.

Yes.

$\epsilon^i{}_{jk} = g^i{}_l \epsilon^l{}_{jk}$

$\epsilon^i{}_{jk}$ is just a linear combo of the same antisymmetric components.

$$\epsilon'^{0123} = \det \Lambda = 1$$

Proper Lorentz

even when j & k are low indices? i & j & k cannot repeat?

$i,j,k = 1,2,3$

$$(\vec{a}\times\vec{b})^i = \epsilon^i{}_{jk} a^j b^k$$

this is a vector because it has 1 free index

$x \quad y \quad z$

$a_x \quad a_y \quad b_z$

$b_x \quad b_y \quad b_z$.

$$(\vec{a}\times\vec{b})_x = \epsilon^i{}_{jk} a^j b^k = a_y b_z - a_z b_y$$

Hence when $i=x$, j & k have to choose from y & z.

I think here we are talking about Euclidean space $= \epsilon^{abc} = \epsilon_{abc}$ 4 dim

$$\epsilon^{abc}\epsilon_{cde} = \delta^a{}_d \delta^b{}_e - \delta^a{}_e \delta^b{}_d$$

gives you either 1 or -1 if $a\neq b$ & $d\neq e$.

$1 \quad -1$

$\epsilon^{123} \quad \epsilon^{213}$

$\epsilon^{321} \quad \epsilon^{312}$

$-1 \quad 1$

$\hookrightarrow$ $\delta^i{}_j$'s are invariant

$\hookrightarrow$ the minus sign is added for antisymmetry.

$$= \underset{1\cdot1}{\epsilon^{123}\epsilon_{312}} + \underset{1\,(-1)}{\epsilon^{123}\epsilon_{321}} + \underset{(-1)\,(-1)}{\epsilon^{213}\epsilon_{321}} + \underset{(-1)\,(1)}{\epsilon^{213}\epsilon_{312}} = 0$$

$$[\vec{a}\times(\vec{b}\times\vec{c})]^i = \epsilon^i{}_{jk}\, a^j\, \epsilon^k{}_{mn}\, b^m c^n$$

derive this

$$= (\delta^i{}_m \delta_{jn} - \delta^i{}_n \delta_{jm})\, a^j b^m c^n$$

$$= b^i(\vec{a}\cdot\vec{c}) - c^i(\vec{a}\cdot\vec{b})$$

$j=n$: dot

$$\vec{a}\times(\vec{b}\times\vec{c}) = \vec{b}(\vec{a}\cdot\vec{c}) - \vec{c}(\vec{a}\cdot\vec{b})$$ ⇒ cool!

$$(S^{-1})^b{}_a (S)^a{}_b = I$$

$$x'^a = S^a{}_b x^b \Rightarrow \boxed{x^b = (S^{-1})^b{}_a x'^a} \Rightarrow \frac{\partial x^b}{\partial x'^a} = (S^{-1})^b{}_a$$

$$\underbrace{\frac{\partial}{\partial x'^a}}_{} = \frac{\partial x^b}{\partial x'^a}\frac{\partial}{\partial x^b} = (S^{-1})^b{}_a \frac{\partial}{\partial x^b}$$

why the partial is a low index tensor.

matrix form

$$= \underbrace{(S^{-T})_a{}^b}_{} \frac{\partial}{\partial x^b}$$

the $\frac{\partial}{\partial x'^a}$ transforms with $(S^{-T})_a{}^b$, so it's a low index vector.

$$\frac{\partial}{\partial x^a} = \partial_a$$

$$\frac{\partial}{\partial x_a} = \partial^a$$

$$\frac{\partial}{\partial x^a} x^b = \delta^b{}_a \rightarrow$$ how do you transform.

Levi-Civita Tensor

$$\eta^{\mu\rho} = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$V_a U^b \rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ tensor}$$

$$\delta^\mu{}_\nu = \eta^{\mu\rho}\eta_{\rho\nu}$$

$$V_a U^a \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$t^{ab}{}_c t^{de} \rightarrow \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{ tensor}$$

$\rho = 0$

$$\sum_{\rho=1}^{4} \begin{cases} \eta^{00}\eta_{00} = (-1)(-1) = 1 \checkmark \\ \eta^{10}\eta_{01} = (0)(0) = 0 \checkmark \\ \ldots \end{cases}$$

$$\sum_{\rho=1}^{4} \begin{cases} \eta^{11}\eta_{11} = (1)(1) = 1 \\ \eta^{21}\eta_{12} \\ \ldots \end{cases}$$

$$V_\mu U^\mu = V^\mu U_\mu$$

$$V_\mu \underbrace{\delta^\mu{}_\nu} U^\nu = V_\mu \underbrace{\eta^{\mu\rho}\eta_{\rho\nu}} U^\nu$$

$$= \cancel{V^\rho \eta^{\mu\rho}} V^\rho U_\rho$$

$$= V^\mu U_\mu$$